

**FMUSER FUQ-4335X Encoder 4-8-12 Input HD-SDI-HDMI
HEVC-H.265 H.264 IP Output UDP-RTP-RTSP for Broadcast
Systems**



HEVC/H.265 Encoding 1.5/2/2.5/3Mbps Bit Rate TS Download

OSD (Logo/PPT/Caption/QR Code) Insertion



- ◆ **Ultra Low Bit Rate:** Save 75% Bandwidth
- ◆ **Enhance Picture Quality:** Advanced Compressing Algorithm
- ◆ **Advanced Pretreatment** De-interlacing, Noise Reduction, Sharpening



News Channel/Movies
1Mbps Full HD



Sports Channel
2Mbps Full HD



B frame(IBBP) GOP Structure



HDMI 1.4



Full HD 1080P



HDCP 1.4



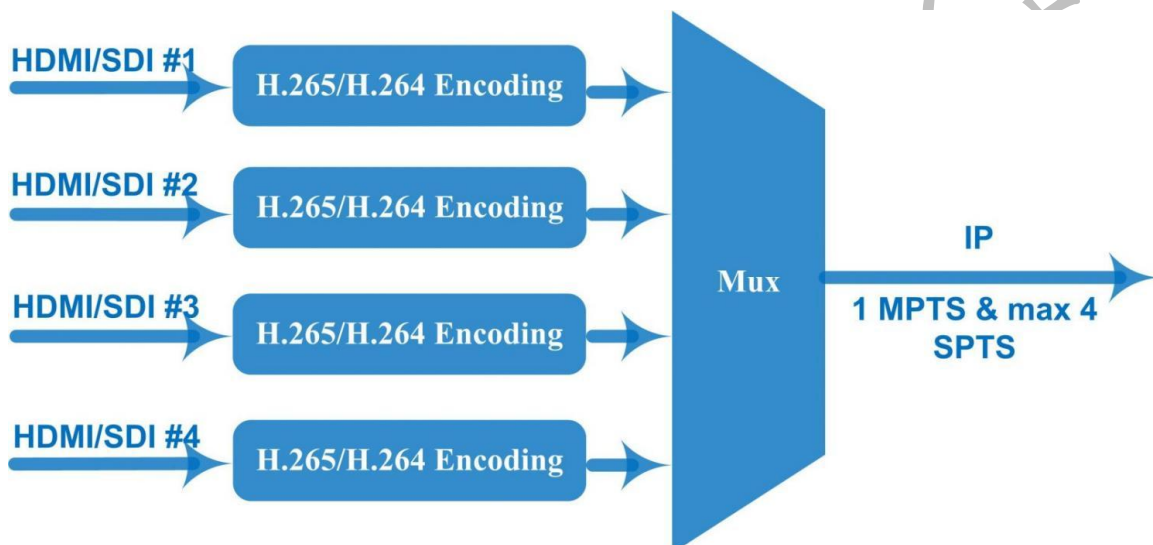
Decoder/STB

STB Available with Ensurity CAS
Decoding Chipset: Montage CS8051/CS8021
NationalChip GX3201H



Up to 2160P 30Hz

Principle Chart of Per Module (the number of encoding channels depend on encoding format and resolution)



Product overview

The FMUSER FUQ-4335X encoder supports HEVC/H.265 and H.264 encoding with inputs including 4/8/12×HD-SDI or HDMI and IP output via UDP/RTP/RTSP. It processes high-definition signals efficiently for transmission. Featuring compact dimensions (482mm×328mm×44mm), lightweight design (5kg), and a wide operating temperature range (0~45℃), it is ideal for broadcast systems and digital headends.

Key feature

Offers robust design: compact dimensions (482mm×328mm×44mm), lightweight (5kg), and wide temperature range (0~45℃ work, -20~80℃ storage).

Technical specification

Input	4/8/12×HD-SDI or HDMI (1.4) inputs for option, HDCP 1.4	4/8/12×HD-SDI or HDMI (1.4) inputs for option, HDCP 1.4	4/8/12×HD-SDI or HDMI (1.4) inputs for option, HDCP 1.4	4/8/12×HD-SDI or HDMI (1.4) inputs for option, HDCP 1.4	4/8/12×HD-SDI or HDMI (1.4) inputs for option, HDCP 1.4
Video Encoding	Encoding Format	HEVC/ H.265, MPEG 4 AVC/H.264	HEVC/ H.265, MPEG 4 AVC/H.264	HEVC/ H.265, MPEG 4 AVC/H.264	HEVC/ H.265, MPEG 4 AVC/H.264
Video Encoding	Resolution (HDMI V1, and SDI)	Input	Input	Output	Output
Video Encoding	Resolution (HDMI V1, and SDI)	Input	Input	HEVC/H.265	MPEG-4 AVC/H.264
Video Encoding	Resolution (HDMI V1, and SDI)	4*1080P-50	4*1080P-50	4*1080P-50	2*1080P-50
Video Encoding	Resolution (HDMI V1, and SDI)	4*1080P-60	4*1080P-60	4*1080P-60	2*1080P-60
Video Encoding	Resolution (HDMI V1, and SDI)	4*1080I-50	4*1080I-50	4*1080P-50	4*1080I-50 2*1080P-50
Video Encoding	Resolution (HDMI V1, and SDI)	4*1080I-60	4*1080I-60	4*1080P-60	4*1080I-60 2*1080P-60
Video Encoding	Resolution (HDMI V1, and SDI)	4*720P-50	4*720P-50	4*720P-50	4*720P-50

Video Encoding	Resolution (HDMI V1, and SDI)	4*720P-60	4*720P-60	4*720P-60	4*720P-60
Video Encoding	Resolution (HDMI V2)	3840×2160_30P, 3840×2160_29.97P (Encoding 2 CHs per module for H.265, and encoding 1 CH for H.264) 1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P, (Encoding 4 CHs per module for H.265, and encoding 2 CHs for H.264) 1280×720_60P, 1280×720_59.94P, 1280×720_50P (Encoding 4 CHs per module for H.264 and H.265)	3840×2160_30P, 3840×2160_29.97P (Encoding 2 CHs per module for H.265, and encoding 1 CH for H.264) 1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P, (Encoding 4 CHs per module for H.265, and encoding 2 CHs for H.264) 1280×720_60P, 1280×720_59.94P, 1280×720_50P (Encoding 4 CHs per module for H.264 and H.265)	3840×2160_30P, 3840×2160_29.97P (Encoding 2 CHs per module for H.265, and encoding 1 CH for H.264) 1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P, (Encoding 4 CHs per module for H.265, and encoding 2 CHs for H.264) 1280×720_60P, 1280×720_59.94P, 1280×720_50P (Encoding 4 CHs per module for H.264 and H.265)	3840×2160_30P, 3840×2160_29.97P (Encoding 2 CHs per module for H.265, and encoding 1 CH for H.264) 1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P, (Encoding 4 CHs per module for H.265, and encoding 2 CHs for H.264) 1280×720_60P, 1280×720_59.94P, 1280×720_50P (Encoding 4 CHs per module for H.264 and H.265)
Video Encoding	Chroma	4:2:0	4:2:0	4:2:0	4:2:0
Video Encoding	Bitrate	0.5Mbps~20Mbps (each channel)	0.5Mbps~20Mbps (each channel)	0.5Mbps~20Mbps (each channel)	0.5Mbps~20Mbps (each channel)
Video Encoding	Rate Control	CBR/VBR	CBR/VBR	CBR/VBR	CBR/VBR
Video Encoding	GOP Structure	IBBP, IPPP	IBBP, IPPP	IBBP, IPPP	IBBP, IPPP

Video Encoding	Advanced Pretreatment	De-interlacing , Noise Reduction, Sharpening	De-interlacing , Noise Reduction, Sharpening	De-interlacing , Noise Reduction, Sharpening	De-interlacing , Noise Reduction, Sharpening
Audio Encoding	Encoding Format	MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough	MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough	MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough	MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough
Audio Encoding	Sampling rate	48KHz	48KHz	48KHz	48KHz
Audio Encoding	Bit-rate (each channel)	48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)	48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)	48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)	48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)
Audio Encoding	Audio Gain	0~255	0~255	0~255	0~255
Stream output	IP (1 MPTS and maximum 4 SPTS) output over UDP/RTP/RTSP per module, 1000M/100M Base-T Ethernet interface (unicast/multicast); IPv4, IPv6 output IP null packet filter	IP (1 MPTS and maximum 4 SPTS) output over UDP/RTP/RTSP per module, 1000M/100M Base-T Ethernet interface (unicast/multicast); IPv4, IPv6 output IP null packet filter	IP (1 MPTS and maximum 4 SPTS) output over UDP/RTP/RTSP per module, 1000M/100M Base-T Ethernet interface (unicast/multicast); IPv4, IPv6 output IP null packet filter	IP (1 MPTS and maximum 4 SPTS) output over UDP/RTP/RTSP per module, 1000M/100M Base-T Ethernet interface (unicast/multicast); IPv4, IPv6 output IP null packet filter	IP (1 MPTS and maximum 4 SPTS) output over UDP/RTP/RTSP per module, 1000M/100M Base-T Ethernet interface (unicast/multicast); IPv4, IPv6 output IP null packet filter
System	Web based management	Web based management	Web based management	Web based management	Web based management

System	Chinese-English control interface	Chinese-English control interface	Chinese-English control interface	Chinese-English control interface	Chinese-English control interface
System	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade
Miscellaneous	Dimension (W× L× H)	Dimension (W× L× H)	482mm×328mm×44mm	482mm×328mm×44mm	482mm×328mm×44mm
Miscellaneous	Approx weight	Approx weight	5kg	5kg	5kg
Miscellaneous	Temperature	Temperature	0~45℃(work), -20~80℃Storage	0~45℃(work), -20~80℃Storage	0~45℃(work), -20~80℃Storage
Miscellaneous	Power	Power	AC 100V-220V±10%, 50/60Hz	AC 100V-220V±10%, 50/60Hz	AC 100V-220V±10%, 50/60Hz

Order Guide

	FU-4335V HDMI V1	FU-4335V HDMI V2	FU-4335V SDI
Input	4/8/12 HDMI	4/8/12 HDMI	4/8/12 SDI
Resolution	1080p, 1080i, 720p	4K-30p, 1080p, 720p	1080p, 1080i, 720p
OSD (logo/QR code) Insertion Function	support	N/A	optional

FMUSER HEVC/H.265 encoder's advantages

1. Providing smooth TS for modulators

FMUSER HEVC/H.265 encoder adopts Fujitsu chip which offers stable bitrate with lower fluctuation compared with other encoding chips, so it provides smooth TS for modulators. It is widely used in variety of digital distribution systems such as CATV digital head-end, satellite and terrestrial digital TV, etc.

2. Encoding with highest compression format—B frame (IBBP)

What is B Frame?

There are 3 major picture types used in the different video algorithms, they are I, P and B.

They are different in the following characteristics:

Iframes are the least compressible but don't require other video frames to decode.

Pframes can use data from previous frames to decompress and are more compressible than Iframes.

Bframes can use both previous and forward frames for data reference to get the highest amount of data compression.

Frame Type	Byte of data/KB	Compression Ratio
------------	-----------------	-------------------

I	18	7:1
P	6	20:1
B	2.5	50:1

In one word, B frame is the highest compression format which makes it possible to process HD video at low bit rate. HEVC/H.265 encoder is not able to save bandwidth unless it is with B frame. In encoder parameters, B frame is often described in GOP (Group of Pictures) structure, like "IBBP".

How to check the GOP structure of the stream from an Encoder?

Please click the link below:

<https://my.pcloud.com/publink/show?code=kZQNH7ZnJBopqRQCchwgS1surdtnmd6cljk>

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

Contact:Tomleequan Email■sales@fmuser.com whatsapp: [+8613922702227](https://wa.me/8613922702227)